

- 3 (a) Fig. 3.1 is a transmission electron micrograph showing two adjacent cells in a leaf.

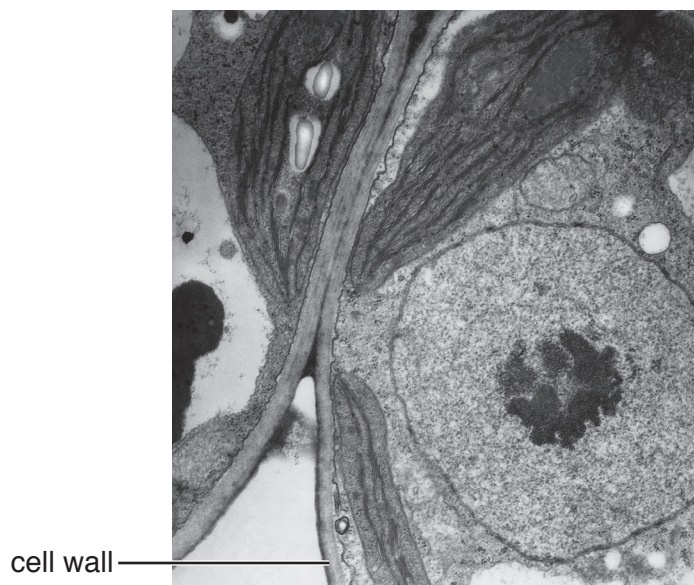


Fig. 3.1

- (i) Cellulose is the main polysaccharide in cell walls of plants.

Describe the structure of cellulose.

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- (ii) State **one** feature **visible** in Fig. 3.1, other than the cell wall, that identifies the cells as plant cells.

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(iii) Outline the role of ATP in a leaf cell.

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Water is a main component of plant cells.

(b) Fig. 3.2 shows two water molecules linked by a hydrogen bond.

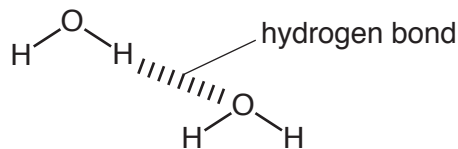


Fig. 3.2

Explain how hydrogen bonding occurs between water molecules.

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(c) Suggest why water is an excellent solvent for ions.

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